

HOG + Logistic Regression

Bottle crack patch classification report

Run: 20261003_153422

Labels: good = 0; crack = 1.

Existing train/valid/test assignments were retained. Exact duplicate image pixels across splits were checked. This check cannot verify physical-bottle separation; that must already be correct in your folders.

Split	Good images	Crack images
train	315	70
valid	105	14
test	32	8

Model and selection procedure

Image resize: (128, 128); grayscale input.

HOG settings: {'orientations': 9, 'pixels_per_cell': (8, 8), 'cells_per_block': (2, 2), 'block_norm': 'L2-Hys', 'transform_sqrt': True, 'feature_vector': True}

HOG feature count: 8100. StandardScaler was fitted on training features only.

Selected C: 0.01; class weight: None; decision threshold: 0.40.

C, class weighting and threshold were selected using validation crack F1. The chosen model remains fitted on train only. Test data were not used for selection.

Training scores are resubstitution scores. Validation scores are selection scores. Test scores provide the held-out evaluation.

Train / validation / test metrics

Metric	Train	Valid	Test
Accuracy	1.0000	0.9916	0.9500
Balanced accuracy	1.0000	0.9952	0.9219
Crack precision	1.0000	0.9333	0.8750
Crack recall	1.0000	1.0000	0.8750
Crack F1	1.0000	0.9655	0.8750
Specificity	1.0000	0.9905	0.9688
ROC AUC	1.0000	0.9993	0.9922
Average precision (AP)	1.0000	0.9952	0.9705

Split	Log loss	TP	TN	FP	FN
train	0.0071	70	315	0	0
valid	0.0323	14	104	1	0
test	0.1217	7	31	1	1

Average precision (AP) summarizes the precision-recall curve; it is not the trapezoidal area under that curve.

These are patch-classification metrics. Whole-bottle inspection requires a separate patch-scanning and aggregation evaluation.

Runtime and saved model

Total tuning time: 6.81 seconds.

Mean patch pipeline time: 10.37 ms over 40 test images.

Timing includes image reading, resizing, HOG extraction, scaling and classification, with warm-up excluded. Drive access and hardware affect these timings.

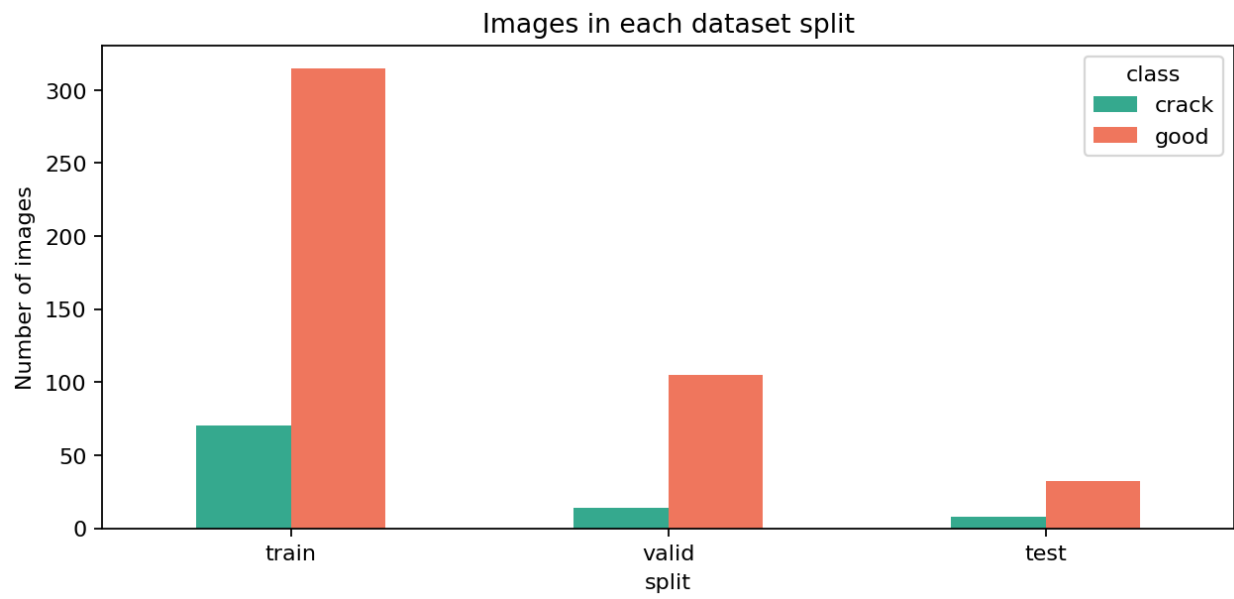
Model size: 0.25 MB.

Python: 3.13.15; scikit-learn: 1.6.1; scikit-image: 0.25.2; Matplotlib: 3.10.0.

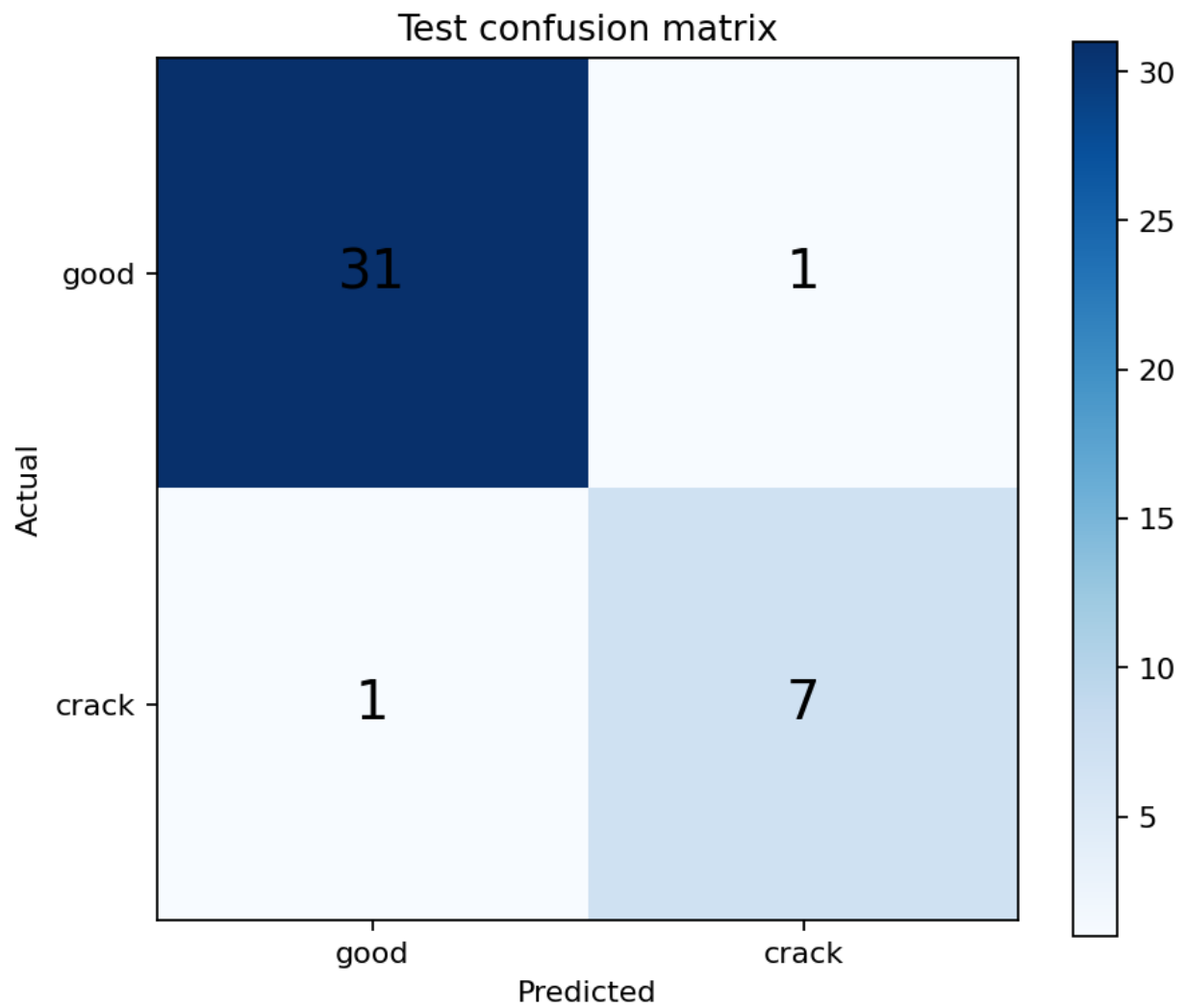
Validation search

C	Weight	Threshold	Valid F1	Converged
0.01	None	0.40	0.9655	True
0.01	balanced	0.50	0.9655	True
0.1	None	0.43	0.9655	True
0.1	balanced	0.50	0.9655	True
1.0	None	0.50	0.9655	True
1.0	balanced	0.50	0.9333	True
10.0	None	0.50	0.9032	True
10.0	balanced	0.82	0.9032	True

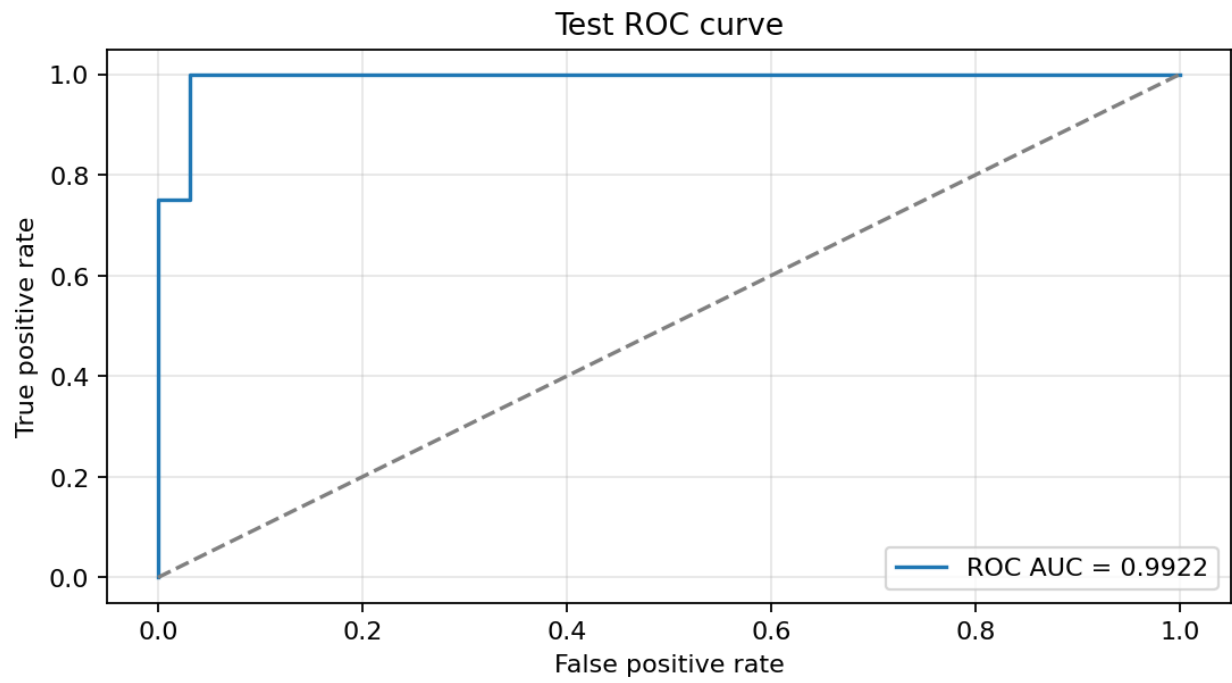
Dataset Distribution



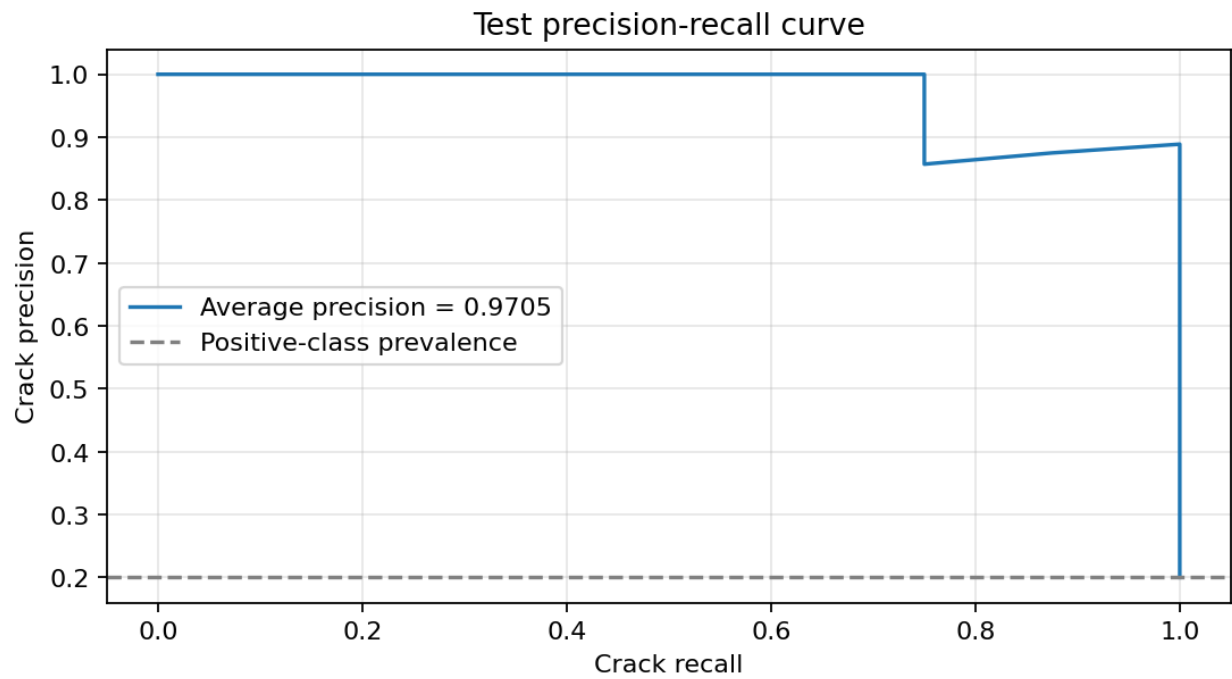
Test Confusion Matrix



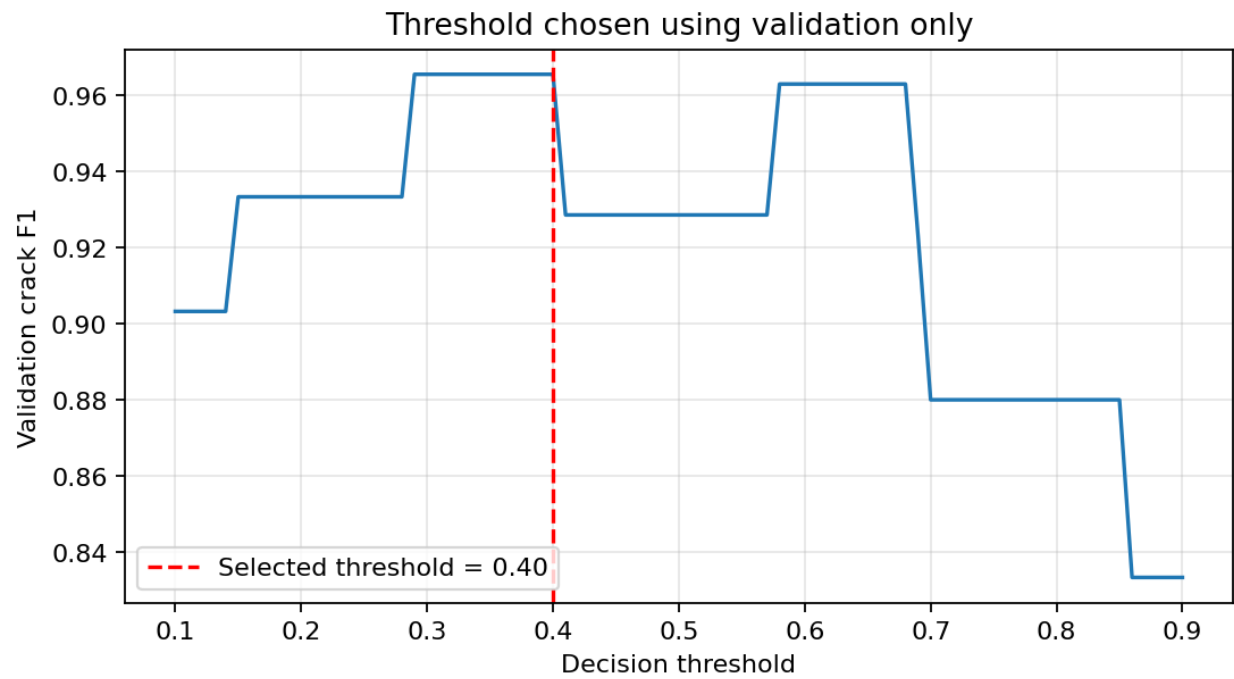
Test Roc



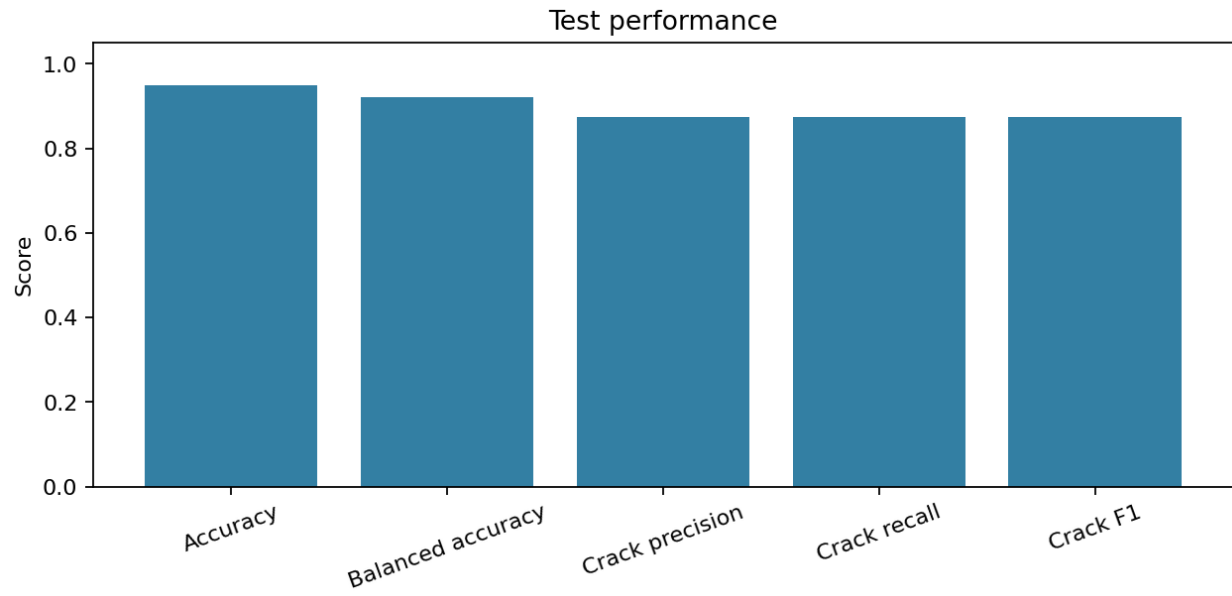
Test Precision Recall



Validation Threshold



Test Metrics



Hog Example

Preprocessed training patch



HOG visualization

